

Python Mini Project Assessment – Ride Sharing System

Time: 1 Hour 30 Minutes

Full Marks: 100

Case Study

A ride-sharing company similar to Pathao or InDrive wants a simple system to manage ride bookings. The company operates different types of vehicles and needs a program that can calculate ride fares, maintain ride history, and ensure proper data validation.

As a Python developer, you are required to develop the system using Object-Oriented Programming concepts and Python best practices.

Project Structure

Students must organize their project using the following module structure:

```
ride_sharing_system/  
|  
├─ main.py  
├─ vehicles.py  
├─ decorators.py  
├─ exceptions.py  
└─ ride_history.txt
```

Module Description

vehicles.py

Contains:

- Abstract class `Vehicle`
- Child classes `Bike` and `Car`
- Fare calculation logic
- Property and setter for vehicle ratings

decorators.py

Contains:

- Custom decorator `ride_logger`

exceptions.py

Contains:

- Custom exception classes (if required)

main.py

Contains:

- User interaction
 - Ride booking functionality
 - Exception handling
 - File operations
 - Program execution
-

Requirements

1. Create an abstract class named `Vehicle` containing common vehicle information and an abstract method for fare calculation.
2. Create at least two vehicle types (`Bike` and `Car`) that inherit from the `Vehicle` class and implement their own fare calculation logic.
3. Use the `@property` decorator to manage and validate vehicle ratings. Ratings must be between 1 and 5.
4. Create a custom decorator that logs a message whenever a ride is successfully booked.
5. Implement exception handling to manage:
6. Invalid ratings
7. Negative ride distances
8. Errors while reading or writing ride history files
9. Organize the project using the module structure provided above.

10. When a ride is booked, store the ride details in a file named `ride_history.txt`.

11. Display ride details including:

12. Driver Name

13. Vehicle Type

14. Distance Travelled

15. Fare Amount

Sample Output

```
Enter Vehicle Type: Car
Enter Driver Name: Ram
Enter Rating: 4.5
Enter Distance: 12

Ride booked successfully!

Driver: Ram
Vehicle: Car
Distance: 12 km
Fare: Rs. 300
```

Instructions

- Use abstraction, inheritance, decorators, properties, modules, and exception handling appropriately.
- Follow proper OOP principles.
- Write clean, readable, and maintainable code.
- Submit all source files along with the generated `ride_history.txt` file.